

SUMMARY STATISTICS

new 1.7–1.8 • old 1.7–1.8

ZONE A · THE FORMULAS

$$\bar{x} = \frac{\sum x_i}{n}$$

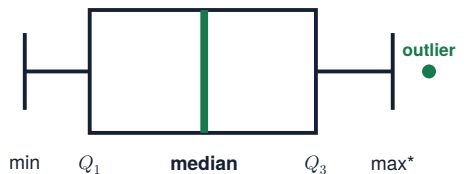
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$\text{IQR} = Q_3 - Q_1$$

outlier if $x > Q_3 + 1.5(\text{IQR})$
or $x < Q_1 - 1.5(\text{IQR})$

A BOX PLOT IS A FIVE-NUMBER STORY



*The whisker stops at the last non-outlier. Plot every outlier separately.

CENTER + SPREAD WORK AS A PAIR

symmetric, no outliers → $\bar{x}$ and s
skewed or outliers → median and IQR

ZONE B · WHAT THE SYMBOLS MEAN

SAMPLE

$\bar{x}$ = sample mean s = sample SD
five-number summary: min, Q_1 , median, Q_3 , max

POPULATION + RESISTANCE

μ = population mean σ = population SD
resistant: median, IQR **not resistant:** mean, SD, range

ZONE C · SAY IT IN WORDS

"Typically, _____ varies from the mean by about _____."

variable in context

value of s

units

ZONE D · WATCH OUT

THE DENOMINATOR

For a **sample** variance or SD, divide by $n - 1$, not n .

TRANSFORMATIONS

Adding a constant shifts the center but not the spread. Multiplying by a constant scales **both**.

CONFIDENCE INTERVALS

master recipe • also used in Units 4 and 5

ZONE A • THE FORMULAS

$$\text{statistic} \pm \underbrace{(\text{critical value})(\text{standard error})}_{\text{margin of error}}$$

$$ME = z^* \cdot SE \quad (\text{proportions})$$

$$ME = t^* \cdot SE \quad (\text{means})$$

$$\text{width} \propto \frac{1}{\sqrt{n}} \Rightarrow \text{halve the margin} \Rightarrow \text{quadruple } n$$

THE FOUR-STEP FRAME

- 1 STATE** Name the parameter in context.
- 2 PLAN** Name the interval; check conditions.
- 3 DO** Formula, substitutions, and interval.
- 4 CONCLUDE** Interpret the interval in context.

COMMON z^* VALUES

confidence	z^*
90%	1.645
95%	1.960
99%	2.576

Get t^* from the table or calculator using **df**.

ZONE B • CONDITIONS — ALWAYS IN THIS ORDER

1 • RANDOM

Random sample or
randomized experiment.

2 • 10%

Sampling without replacement:
 $n < 0.10N$.

3 • NORMAL / LARGE COUNTS

proportions: $n\hat{p} \geq 10$, $n(1 - \hat{p}) \geq 10$
means: normal population, $n \geq 30$, or no strong
skew/outliers

ZONE C • SAY IT IN WORDS

"We are _____% confident that the interval from _____ to _____ captures the true _____."

Confidence level: If we took many samples and built an interval each time,
about _____% of those intervals would capture the true _____.

ZONE D • WATCH OUT

THE INTERVAL IS ABOUT THIS SAMPLE. THE LEVEL IS ABOUT THE METHOD.

Never say the fixed parameter has a 95% chance of being inside the interval. After the interval is calculated, it either captures the parameter or it does not.

RESIDUALS, r^2 & MODEL FIT

new 5.4–5.5 • old 2.7–2.8 • ★ old 2.9, 9.1–9.5

ZONE A · THE FORMULAS + THE PLOTS

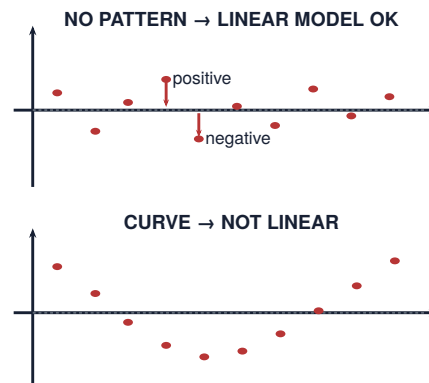
$$\text{residual} = y - \hat{y} = \text{actual} - \text{predicted}$$

r^2 = fraction of the variation in y explained by the linear relationship with x

$$s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

s is the **typical size of a residual**. It has the same units as the response variable.

READ THE RESIDUAL PLOT



ZONE B · COMPUTER-OUTPUT DECODER

Predictor	Coef	SE Coef	T	P	Read it as
Constant	12.41	1.86	6.67	<.001	→ a , the intercept
Study hours	4.83	0.72	6.71	<.001	→ b , s_b , t , P -value
$S = 3.17$		→ s	$R\text{-Sq} = 78.4\% \rightarrow r^2$		

ZONE C · SAY IT IN WORDS

"About _____% of the variation in _____ is explained by the linear relationship with _____."

"The actual _____ was _____ units _____ what the line predicted."

ZONE D · WATCH OUT + BEYOND

MODEL-FIT RULES

No extrapolation: do not predict beyond the observed x -range.

r^2 gives the explained fraction, not the direction. Use r for direction.

★ BEYOND THE EXAM

Departures: outlier · high leverage · influential point

$$\log(\hat{y}) = a + bx \Rightarrow \hat{y} = 10^{a+bx}$$

Slope inference ($df = n - 2$):

$$\mu_b = \beta \quad \sigma_b = \frac{s}{\sigma_x \sqrt{n}}$$

$$s_b = \frac{s}{s_x \sqrt{n-1}} \quad t = \frac{b}{s_b} \quad b \pm t^* s_b$$